ASDJ
AINSHAMS DENTAL JOURNAL

Print ISSN 1110-7642
Online ISSN 2735-5039

AIN SHAMS DENTAL JOURNAL
Official Publication of Ain Shams Dental School
September 2021 • Vol. XXIV

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Introduction

Pain is an unpleasant sensation that is attended by actual or potential tissue damage and is the body's response to noxious stimulus. Postoperative pain is usually due to an acute inflammatory response in the periradicular tissue, provoked by extrusion of debris and irrigating solutions beyond the apical constriction, which causes severe tissue irritation and necrosis. Irrigating solutions are supposed to act as lubricants and as cleaning agents during biomechanical endodontic treatment. They must be in contact with walls of the canal to ensure effective action, especially in the apical third1.

Even with good instrumentation, total removal of debris of canals is reported to be difficult. Most of root canal walls remain untouched, which is confirmed with micro computed tomography scanning 2. These untouched areas are thought to be a source of reinfection or persistent periradicular inflammation3 because they may harbor debris of pulpal tissues, bacteria, and their byproducts. This emphasizes the need for another means of disinfecting all of the root canal. Mechanical active irrigation reduces post-operative pain4. It was shown to improve canal and isthmus cleanliness and to reduce postoperative pain. The bacterial number was more with active methods of irrigation. It may be concluded that they are effective in decreasing pain levels after treatment and improving canals cleanliness during Endo treatment 4. The simplest way to activate the irrigating solution inside the root canal is by using manual dynamic agitation technique. It is considered simple and cost-effective, but some drawbacks hinder its clinical use. It was reported that the irrigant was delivered 1 mm deeper than the tip of the needle, thus showing difficulty in accessing the apical part of the canal 5.

Up to our knowledge, no study compared 6 hours, 24 hours ,48hours time intervals together clinically. Thus, this study was to make a comparison between levels of pain after manual dynamic activation of irrigation. Our null hypothesis was that there is no difference in the level of post operative pain after different time intervals after manual dynamic agitation.

Materials and Methods:

- ***Patients' selection***

The protocol of the trial was approved by the Ethics committe, Faculty of Dentistry, Ain Shams University, Cairo, Egypt. The inclusion criteria were based on female patients, medically-free of any systemic disease, patients with an age range of 25-45 years, symptomatic irreversible pulpitis in lower molars with mature roots, teeth without calcified root canals, root caries-free teeth (Restorable teeth), teeth without internal or external root resorption, While the exclusion criteria were based on pregnant females, male patients,patients having bruxism or clenching, previously endodontically treated teeth, periodontally affected teeth, presence of abnormal mobility, presence of resorption, periapical abscess, periodontal space widening or root fracture, and teeth without internal or external root resorption.

- ***Procedural steps***

Personal information includes the participant's name, phone number, address medical and dental history. Chief complaint and history of pain were recorded in the patient's own words. Extra oral examination was performed to rule out swelling or facial cellulitis. Intraoral clinical examination was performed visually for the presence of large or old restorations, extensive caries, or traumatic injuries, as well as percussion and palpation to rule out apical periodontitis. Clinical diagnosis of acute pulpitis was detected by a prolonged exaggerated response> 10 seconds. The Radiographic examination was done using intra-oral periapical films with a bisecting angle technique.

- ***Treatment procedures***

22 Patients were selected from the postgraduate outpatient clinic in the Endo department, Faculty of Dentistry, Ain Shams University,cairo,Egypt. A nerve block of local anesthesia was used to anaesthetize the tooth (Art pharma anthesia, ArtPharma Pharmaceuticals, 6 October City, Egypt). Access was done using z and round burr(Dentsply Tulsa Dental Specialties; Tulsa, Oklahoma, USA). Isolation was carried out with a rubber dam. Length was determined using apex locator (E connect apex locator, Eighteeth, Jiangsu Province, China) and confirmed by periapical radiograph. Cleaning and shaping of root canals was done by the crown down tech using Fanta rotary files using an endo motor with speed and torque following the manufacturer's instructions (Fanta Dental Materials Co., Ltd, East Tianlin RD, 200230 Shanghai), by beginning with orifice opener then 20 then 25 rotary file then file 30 till 35 in root with multiple canals and till 50 if single canal in root . Irrigation and Final irrigation protocol: 5 ml of 2.5% Sodium hypochlorite was delivered into the canal using a syringe with end-perforated beveled needle gauge 27. The irrigant was manually activated for 60 seconds with a master cone for each canal. The canals were dried with sterile paper points corresponding to the same size of the master cone. Root canals were obturated by using lateral compaction by gutta percha master cone equal to the size of the master apical file(50 for single canal in root ,35 for more than one canal in root) and a spreader was used with size that shorter 2 ml from working length ,to make space for auxiliary cones (smaller than spreader (size 25)), using resin root canal sealer then placing temporary filling. The patient was asked to come back to complete the treatment procedures.

- ***Pain assessment***

Pain was assessed by giving the patient the numerical pain scale to assess pain at 6, 24, 48 hours after the visit. If swelling was recorded, the patient was instructed to contact the operator to assess the severity and determine if antibiotics or drainage were indicated. NRS (Numerical Rate Scale) 0: no pain, from 1 to 3: mild (recognizable, not discomforting) ,from 4 to 6: moderate (discomforting, but bearable) ,from 7 to 10: severe (considerable discomfort, difficult to bear).

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